

```
install numpy
```

```
!pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
```

```
check version
```

```
import numpy as np  
print(np.__version__)
```

```
2.0.2
```

```
creating multiple types of arrays
```

```
#1D Array  
import numpy as np  
a=np.array([1,2,3,4,5])  
print(a)
```

```
[1 2 3 4 5]
```

```
#2D Array  
b=np.array([[1,2,3],  
            [4,5,6],  
            [7,8,9]  
])  
  
print(b)
```

```
[[1 2 3]  
 [4 5 6]  
 [7 8 9]]
```

```
#3D Array  
c=np.array([  
    [  
        [1,2,3],  
        [4,5,6]  
    ],  
    [  
        [7,8,9],  
        [10,11,12]  
    ],  
    [  
        [13,14,15],  
        [16,17,18]  
    ]  
])
```

```
    ]

    ])
    print(c)
```

```
[[[ 1  2  3]
   [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]

 [[13 14 15]
  [16 17 18]]]
```

Built-in Array Creation Functions

```
import numpy as np
d=np.zeros((3,3))
print(d)
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
d=np.ones((3,3))
print(d)
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
d=np.full((3,3),8)
print(d)
```

```
[[8 8 8]
 [8 8 8]
 [8 8 8]]
```

```
def greet():
    """
    this is not testing
    """
    print("testing function")
print(greet())
print(greet.__doc__)
```

```
testing function
None
```

```
    this is not testing
```

```
#identity matrix
e=np.eye(4)
```

```
print(e)
```

```
[[1. 0. 0. 0.]  
 [0. 1. 0. 0.]  
 [0. 0. 1. 0.]  
 [0. 0. 0. 1.]]
```

```
a=np.ones([2,3])  
b=np.zeros([2,3])  
c=np.full([2,3],9)  
e=np.eye(4)
```

```
print(a,b,c,e)
```

```
[[1. 1. 1.]  
 [1. 1. 1.]] [[0. 0. 0.]  
 [0. 0. 0.]] [[9 9 9]  
 [9 9 9]] [[1. 0. 0. 0.]  
 [0. 1. 0. 0.]  
 [0. 0. 1. 0.]  
 [0. 0. 0. 1.]]
```

```
#Ranges  
f=np.arange(2,10,3)  
print(f)
```

```
[2 5 8]
```

```
np.linspace(0,2,5)
```

```
array([0. , 0.5, 1. , 1.5, 2. ])
```

Random Arrays

```
np.random.randn(3,3)
```

```
array([[ -0.73994095, -0.1740282 ,  0.42055788],  
       [-0.06539609, -0.36525583, -0.28618895],  
       [-1.93268956, -0.89527018, -0.92086157]])
```

```
np.random.rand(3,3)
```

```
array([[0.34895402, 0.71853557, 0.4086203 ],  
       [0.26147915, 0.31839154, 0.20244552],  
       [0.2723554 , 0.18211209, 0.9332443 ]])
```

```
np.random.randint(1,5,size=(4,4))
```

```
array([[3, 4, 1, 3],  
       [2, 2, 3, 4],
```

```
[3, 2, 1, 1],  
[4, 1, 2, 1]])
```

```
np.empty((2,6))
```

```
array([[7.27931619e-319, 0.00000000e+000, 6.61168049e-310,  
        6.61168049e-310, 6.61168039e-310, 6.61168039e-310],  
       [5.36201145e-317, 6.61168080e-310, 6.61168117e-310,  
        6.61170696e-310, 6.61168039e-310, 6.61168039e-310]])
```

Array Attributes

```
arr=np.array([["a","b","c","d","e"], [1,2,3,4,5]])  
print(arr.shape)
```

```
(2, 5)
```

```
print(arr.ndim)
```

```
2
```

```
print(arr.size)
```

```
10
```

```
print(arr.dtype)
```

```
<U21
```

```
print(arr.itemsize)
```

```
84
```

```
print(arr.nbytes)
```

```
840
```

```
arr=np.array([["a","b","c","d","e"], [1,2,3,4,5]])  
print(arr.shape)  
print(arr.ndim)  
print(arr.size)  
print(arr.dtype)  
print(arr.itemsize)  
print(arr.nbytes)
```

```
(2, 5)
```

```
2
```

```
10
```

```
<U21
```

```
84
```

```
840
```

Indexing and Slicing

```
a=np.array([10,20,30,40,50])  
print(a[0])  
print(a[2])  
print(a[-1])
```

```
10  
30  
50
```

```
arr=np.array([1,2,3,4,5,56,78])  
print(arr[1:5])
```

```
[2 3 4 5]
```

```
print(arr[:3])
```

```
[1 2 3]
```

```
print(arr[::2])
```

```
[ 1  3  5 78]
```

```
print(arr[::-1])
```

```
[78 56  5  4  3  2  1]
```

```
m=np.array([[1,2,3],  
            [4,5,6],  
            [7,8,9]])  
print(m[0,0])  
print(m[1,2])
```

```
1  
6
```

```
#Row slicing  
print(m[0,:])
```

```
[1 2 3]
```

```
#Column Slicing  
print(m[:,1])
```

```
[2 5 8]
```

```
#Sub Matrix  
print(m[0:2,1:3])
```

```
[[2 3]
 [5 6]]
```

```
#Boolean Mask
a=np.array([10,25,8,42,17])
mask=a>15
print(a[mask])
```

```
[25 42 17]
```

```
#Shorthand
print(a[a>15])
```

```
[25 42 17]
```

```
#Fancy Indexing
print(a[[0,3]])
```

```
[10 42]
```

Array Operations

```
import numpy as np
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
```

```
[5 7 9]
```

```
print(a-b)
```

```
[-3 -3 -3]
```

```
print(a/b)
```

```
[0.25 0.4 0.5 ]
```

```
print(a*b)
```

```
[ 4 10 18]
```

```
print(a**2)
```

```
[1 4 9]
```

```
print(a+10)
```

```
[11 12 13]
```

```
import numpy as np
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
print(a-b)
print(a/b)
print(a*b)
print(a**2)
print(a+10)
```

```
[5 7 9]
[-3 -3 -3]
[0.25 0.4 0.5 ]
[ 4 10 18]
[1 4 9]
[11 12 13]
```

Mathematical Functions

```
a=np.array([1.0,2,4.5,8.6])
np.floor(a)
```

```
array([1., 2., 4., 8.])
```

```
np.ceil(a)
```

```
array([1., 2., 5., 9.])
```

```
np.round(a,2)
```

```
array([1. , 2. , 4.5, 8.6])
```

```
#Aggregate Functions
m=np.array([[1,2,3],
            [4,5,6]])
```

```
np.sum(m)
```

```
np.int64(21)
```

```
np.mean(m)
```

```
np.float64(3.5)
```

```
np.min(m)
```

```
np.int64(1)
```

```
np.max(m)
```

```
np.int64(6)
```